Statistical Learning and Machine Learning

Exercise Session (Week 10)

SOLUTIONS

BOOK EXERCISES

Problem 5.2

Let $\mathbf{x}_n \in \mathbb{R}^D$, $t_n \in \mathbb{R}$, and $\{(\mathbf{x}_n, t_n)\}_{n=1}^N$ denote our observations. The likelihood function of (5.16) is given by:

$$\mathcal{L}\left(\mathbf{t}|\mathbf{X}, \mathbf{w}, \beta\right) = \prod_{n=1}^{N} p\left(t_n|\mathbf{x}_n, \mathbf{w}\right) = \prod_{n=1}^{N} \mathcal{N}\left(t_n|\mathbf{y}(\mathbf{x}_n, \mathbf{w}), \beta^{-1}\mathbf{I}\right)$$

where $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ and $\mathbf{t} = \{t_1, \ldots, t_n\}$. The log likelihood function of (5.16) is then given by:

$$\ln\left(\mathcal{L}\left(\mathbf{t}|\mathbf{X}, \mathbf{w}, \beta\right)\right) = \ln\left(\prod_{n=1}^{N} \mathcal{N}\left(t_n|\mathbf{y}(\mathbf{x}_n, \mathbf{w}), \beta^{-1}\mathbf{I}\right)\right) \tag{1}$$

$$= \sum_{n=1}^{N} \ln\left(\mathcal{N}\left(t_n|\mathbf{y}(\mathbf{x}_n, \mathbf{w}), \beta^{-1}\mathbf{I}\right)\right) \tag{2}$$

$$= \sum_{n=1}^{N} \ln\left(\frac{1}{(2\pi)^{D/2}} \frac{1}{|\beta^{-1}\mathbf{I}|^{1/2}} \exp\left\{\frac{-1}{2}\left(t_n - \mathbf{y}(\mathbf{x}_n, \mathbf{w})\right)^T \left(\beta^{-1}\mathbf{I}\right)^{-1} \left(t_n - \mathbf{y}(\mathbf{x}_n, \mathbf{w})\right)\right\}\right) \tag{3}$$

$$= \frac{-1}{2} \sum_{n=1}^{N} \left(t_n - \mathbf{y}(\mathbf{x}_n, \mathbf{w})\right)^T (\beta\mathbf{I}) \left(t_n - \mathbf{y}(\mathbf{x}_n, \mathbf{w})\right) + \underbrace{\sum_{n=1}^{N} \ln\left(\frac{1}{(2\pi)^{D/2}} \frac{1}{|\beta^{-1}\mathbf{I}|^{1/2}}\right)}_{\text{independent of } \mathbf{w}} \tag{4}$$

$$= \frac{-\beta}{2} \sum_{n=1}^{N} \|t_n - \mathbf{y}(\mathbf{x}_n, \mathbf{w})\|^2 + \underbrace{\sum_{n=1}^{N} \ln\left(\frac{1}{(2\pi)^{D/2}} \frac{1}{|\beta^{-1}\mathbf{I}|^{1/2}}\right)}_{\text{independent of } \mathbf{w}} \tag{5}$$

The first term in equation (5) is negative proportional to (5.11). The second term in equation (5) is independent of $\mathbf{w}$, and therefore it does not effect the value of $\mathbf{w}$, when finding the optimal value of $\mathbf{w}$. It only acts as a offset of the value of the log likelihood function. Hence, maximizing Equation (5) w.r.t. $\mathbf{w}$ corresponds to minimizing (5.11) w.r.t. $\mathbf{w}$.

Problem 5.5

Let K be the number of classes of the the multiclass neural network model, N be the number of data points $\mathbf{x} = \{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$, and $t_{n,k} \in \{0, 1\}$ be the binary target variables associated with the output. Given the network output interpretation $y_k(\mathbf{x}_n, \mathbf{w}_k) = p(t_{n,k} = 1 \mid \mathbf{x}_n)$, the conditional distribution of the target vector $\mathbf{t}_n = \{t_{n,1}, \ldots, t_{n,k}\}$ of the nth data point $\mathbf{x}_n$ becomes:

$$p(\mathbf{t}_n \mid \mathbf{w}_1, \ldots, \mathbf{w}_K) = \prod_{k=1}^{K} p(t_{n,k} = 1 \mid \mathbf{x}_n)^{t_{n,k}} = \prod_{k=1}^{K} y_k(\mathbf{x}_n, \mathbf{w}_k)^{t_{n,k}}$$

The likelihood function across all datapoints, then becomes:

$$p\left(\mathbf{T} \mid \mathbf{w}_1, \ldots, \mathbf{w}_K\right) = \prod_{n=1}^{N} \prod_{k=1}^{K} y_k(\mathbf{x}_n, \mathbf{w}_k)^{t_{n,k}}$$

The log likelihood function becomes:

$$\ln\left(p\left(\mathbf{T} \mid \mathbf{w}_1, \ldots, \mathbf{w}_K\right)\right) = \ln\left(\prod_{n=1}^{N}\prod_{k=1}^{K} y_k(\mathbf{x}_n, \mathbf{w}_k)^{t_{n,k}}\right) \tag{6}$$

$$= \sum_{n=1}^{N}\sum_{k=1}^{K} t_{n,k} \ln\left(y_k(\mathbf{x}_n, \mathbf{w}_k)\right) \tag{7}$$

Thus, maximizing (7) is equivalent to minimizing the cross-entropy error function in (5.24).

Problem 5.6

Note: This solution uses (5.23) instead of (5.21), as the former assumes K outputs, whereas the latter only assumes 1 output.
Recap: The logistic sigmoid (4.59) and its derivative (4.88) are given by $\sigma(x) = \frac{1}{1+\exp(-x)}$ and $\frac{d\sigma(x)}{dx} = \sigma(x)(1-\sigma(x))$.
Let $y_{n,k} = \sigma(a_k)$, the taking the derivative of (5.23) with respect to the kth activation a_k:

$$\frac{dE(\mathbf{w})}{da_k} = \frac{d}{da_k}\left\{-\sum_{n=1}^{N}\sum_{c=1}^{K} t_{n,c}\ln\left(y_{n,c}\right) + \left(1-t_{n,c}\right)\ln\left(1-y_{n,c}\right)\right\} \tag{8}$$

$$= -\sum_{n=1}^{N}\sum_{c=1}^{K}\frac{d}{da_k}\left\{t_{n,c}\ln\left(y_{n,c}\right) + \left(1-t_{n,c}\right)\ln\left(1-y_{n,c}\right)\right\} \tag{9}$$

$$= -\sum_{n=1}^{N}\frac{d}{da_k}\left\{t_{n,k}\ln\left(y_{n,k}\right) + \left(1-t_{n,k}\right)\ln\left(1-y_{n,k}\right)\right\} \tag{10}$$

$$= -\sum_{n=1}^{N}\left(t_{n,k}\frac{1}{y_n,k}\frac{d}{da_k}\left\{y_{n,k}\right\} + \left(1-t_{n,k}\right)\frac{1}{1-y_n,k}\frac{d}{da_k}\left\{1-y_{n,k}\right\}\right) \tag{11}$$

$$= -\sum_{n=1}^{N}\left(t_{n,k}\frac{1}{y_n,k}\frac{d}{da_k}\left\{y_{n,k}\right\} + \left(1-t_{n,k}\right)\frac{1}{1-y_n,k}\left(\frac{d}{da_k}\left\{1\right\} - \frac{d}{da_k}\left\{y_{n,k}\right\}\right)\right) \tag{12}$$

$$= -\sum_{n=1}^{N}\left(t_{n,k}\frac{1}{y_{n,k}}y_{n,k}(1-y_{n,k}) + \left(1-t_{n,k}\right)\frac{1}{1-y_{n,k}}\left(0 - y_{n,k}(1-y_{n,k})\right)\right) \tag{13}$$

$$= -\sum_{n=1}^{N}\left(t_{n,k}(1-y_{n,k}) - (1-t_{n,k})y_{n,k}\right) \tag{14}$$

$$= -\sum_{n=1}^{N}\left(t_{n,k} - t_{n,k}y_{n,k} - y_{n,k} + t_{n,k}y_{n,k}\right) \tag{15}$$

$$= -\sum_{n=1}^{N}\left(t_{n,k} - y_{n,k}\right) \tag{16}$$

$$= \sum_{n=1}^{N}\left(y_{n,k} - t_{n,k}\right) \tag{17}$$

Hence, the derivative of (5.23) w.r.t. the activation a_k satisfies (5.18) $\frac{dE}{da_k} = y_k - t_k$.

Notes:

- In (10), we used $\frac{d}{da_k}\{y_c\} = 0$ for $k \neq c$.
- In (11), we used the chain rule to find the derivative of $\ln(y_k(a_k))$ w.r.t. a_k.
- In (13), we used the derivative of the logistic sigmoid (4.88).
- In (14) to (16), we are just simplifying the expression.
- In (17), we move the $-$ from outside the sum to inside the sum.

NEURAL NETWORKS

1) Parameters of the neural network are the weights and the biases of its layers $\mathbf{W}_1, \mathbf{W}_2, \mathbf{W}_3, \mathbf{W}_{out}, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_{out}$. Hyper-parameters are the type of activation functions (and their parameters, if they have) used for each layer, the network architecture.
2) See Fig. 1.

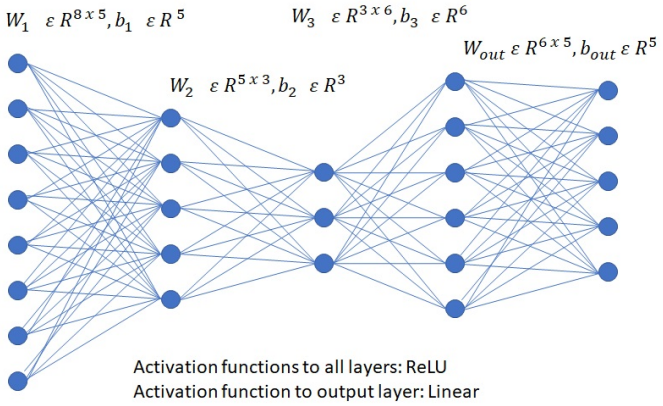

Fig. 1. An MLP mapping 8-d input to 5-d output

3) The hyper-parameter values are selected based on the validation error using different neural network realizations (model selection). On each experiment (for a specific network realization) the parameters of the network are estimated by training it using the Backpropagantion algorithm to minimize the mean-squared-error between the network's outputs and the target vectors.
4) We use the mean-squared-error between the network's outputs and the target vectors. We choose MSE loss because this corresponds to the MLE solution for the network's parameters.
5) Because then we have $\mathbf{W} = \mathbf{W}_1\mathbf{W}_2\mathbf{W}_3\mathbf{W}_{out}$, thus we end up to have a linear regression model (after training). Draw two layers (input with 8 neurons and output with 5 neurons) with connections between them (matrix 8 x 5 plus bias) and linear activation function.
6) We would need to (again) use the Backpropagation algorithm to perform iterative optimization. This is because there are many combinations of $\mathbf{W}_k$'s that can lead to the final $\mathbf{W}$ of the two-layer network.
7) It would probably be inferior, as optimizing directly for $\mathbf{W}$ using linear regression leads to a global optimum of the MSE criterion. Optimizing with Backpropagation for all $\mathbf{W}_k$'s would lead to one choice for the final $\mathbf{W}$ which may not be the optimal.